

Defective Analysis & the Numbers

A Generator-Presentation Derivation of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$

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Claim boundary. This paper uses ordinary mathematical language: generator, action, quotient, kernel, group completion, localization, valuation, absolute value, metric completion, complexification, universal property, and presentation choice. The symbols $\theta, \alpha, \lambda, \rho$ are local notation for standard constructions and obstructions. The terminal claim about $\mathbb{C}$ is stated as a postulate with explicit proof obligations, not as a completed theorem.

Abstract

This note reconstructs the classical scalar ladder

$$0 \longrightarrow \{0, 1\} \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}$$

from a minimal generating presentation. The natural numbers arise from the forward successor action. The integers arise by group completion. The rationals arise by localization at nonzero integers. The reals arise by passing from presentation-dependent load to gauge-invariant size: invariant sizes on $\mathbb{Q}$ are absolute values; Ostrowski's classification and a generator-visibility axiom select the Archimedean absolute value; $\mathbb{Q}$ is incomplete for the selected uniformity; the completion is therefore forced and unique. The complex numbers arise by adjoining a polar unit so that exponentiation admits magnitude-preserving rotation rather than only real scaling. The final postulate is that, under the stated classical hypotheses, $\mathbb{C}$ is the terminal scalar layer obtained by adjoining additive inverses, multiplicative inverses, generator-visible Archimedean completion, and polar complexification.

1. Notation and public-language replacement

Let

$$S_{\min} = \{0, 1\}$$

be the minimal generating set. The symbol 0 denotes the additive identity; 1 denotes the multiplicative identity and the additive generator.

This paper uses four symbols as compact labels:

θ = the additive successor/index action,

α = the multiplicative action generated by repeated addition,

λ = the completion obstruction selected by gauge-invariant, generator-visible size,

ρ = the polarization obstruction exposed by real exponentiation.

These labels are not new mathematical structures. They mark standard transitions in the scalar ladder:

$$\boxed{\mathbb{N} = (\theta^+, \alpha^+), \quad \mathbb{Z} = (\theta^\pm, \alpha^+), \quad \mathbb{Q} = (\theta^\pm, \alpha^\pm), \quad \mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda), \quad \mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).}$$

Here $+$ means only the forward operation is available; $\pm$ means the corresponding inverse operation has been adjoined.

Guiding principle. Each new number system is obtained by making a previously partial operation total, or by passing to a quotient/completion that removes presentation-dependence while preserving the intended scalar operations.

2. The zero stage and the minimal generating set

The construction begins with the distinction between no additive content and a unit generator:

$$S_{\min} = \{0, 1\}.$$

This is the smallest presentation from which the ordinary counting structure can be generated. The element 0 is neutral for addition; the element 1 generates the successor chain.

The set

$$S_2 = \{0, 1, 2\}$$

is not taken as primitive. It is the first successor-closed extension of $S_{\min}$ because

$$2 = 1 + 1.$$

This distinction will matter at the real-number stage: treating 2 as generated rather than primitive creates an apparent dyadic load. That load will be shown to be presentation-dependent rather than scalar-invariant.

3. The natural numbers: forward additive generation

Definition 3.1 (Natural-number presentation). The natural numbers are generated from $S_{\min}$ by the forward successor action

$$\theta(n) = n + 1.$$

Equivalently,

$$\mathbb{N} = \{0, 1, \theta(1), \theta^2(1), \dots\} = \{0, 1, 2, 3, \dots\}.$$

The additive structure is the free commutative monoid on one generator:

$$(\mathbb{N}, +, 0) = \langle 1 \rangle_{\text{cmon}}.$$

Multiplication is induced by repeated addition:

$$mn = \underbrace{n + n + \dots + n}_{m \text{ times}}.$$

Thus

$$(\mathbb{N}, +, \cdot, 0, 1)$$

is a commutative semiring.

The obstruction visible at this stage is that addition is not reversible. For $n > 0$, the equation

$$n + x = 0$$

has no solution in $\mathbb{N}$.

Hence the next scalar system is forced if additive cancellation is to become internal.

4. The integers: group completion

Definition 4.1 (Grothendieck group completion of $\mathbb{N}$). Define

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim$$

where

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

The class of (a, b) is interpreted as the formal difference

$$[(a, b)] = a - b.$$

Then

$$n = [(n, 0)], \quad -n = [(0, n)].$$

Addition and multiplication are defined by

$$[(a, b)] + [(c, d)] = [(a + c, b + d)]$$

and

$$[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)].$$

These operations are well-defined on equivalence classes.

The result is the commutative ring

$$(\mathbb{Z}, +, \cdot, 0, 1).$$

The positive successor action has become an additive group action:

$$\theta^+ \rightsquigarrow \theta^\pm.$$

In the compact notation,

$$\boxed{\mathbb{Z} = (\theta^\pm, \alpha^+)}.$$

The obstruction visible at this stage is that multiplication is not reversible for non-units. For example,

$$2x = 1$$

has no solution in $\mathbb{Z}$.

Thus the next scalar system is forced if every nonzero scalar is to have a multiplicative inverse.

5. The rationals: localization by nonzero integers

Definition 5.1 (Localization of $\mathbb{Z}$). Let

$$S = \mathbb{Z} \setminus \{0\}.$$

The rational numbers are the localization

$$\mathbb{Q} = S^{-1}\mathbb{Z}.$$

Equivalently,

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) / \sim$$

where

$$(a, b) \sim (c, d) \iff ad = bc.$$

The class of (a, b) is written

$$\frac{a}{b}.$$

The result is the field

$$(\mathbb{Q}, +, \cdot, 0, 1).$$

Every nonzero element is invertible:

$$q \neq 0 \implies q^{-1} \in \mathbb{Q}.$$

The multiplicative action has become reversible:

$$\alpha^+ \rightsquigarrow \alpha^\pm.$$

Thus

$$\boxed{\mathbb{Q} = (\theta^\pm, \alpha^\pm)}.$$

At this stage, the ordinary algebraic inverse problem has been solved. The next obstruction is not a missing inverse. It is the problem of extracting a scalar notion of closeness from presentation-dependent generation data.

6. The completion obstruction: gauge variance of presentation load

The rational numbers carry the usual absolute-value metric $d(x, y) = |x - y|$. A tempting argument observes that this metric does not control the complexity of generating a rational from a chosen presentation, and then passes directly to the Cauchy completion. That passage is unforced: completing $\mathbb{Q}$ in the metric $|\cdot|$ does not remove the disagreement between presentation loads, and the privileged status of $|\cdot|$ among possible metrics must be derived rather than assumed. This section closes the gap. The corrected logic is:

1. presentation loads carry no gauge-invariant pointwise content (Lemma 6.3);
2. therefore any notion of scalar closeness must be sourced in the operations, i.e. in a gauge-invariant size function, and the invariance axioms force such a function to be an absolute value (Lemma 6.7);
3. Ostrowski's classification together with one axiom native to this paper – the generating action θ must remain metrically visible – selects the Archimedean absolute value uniquely up to uniform equivalence (Theorem 6.11);
4. $\mathbb{Q}$ is incomplete for that unique invariant uniformity (Proposition 6.13), and the completion is forced and unique (Theorem 7.1).

This is the obstruction λ , now stated as a theorem chain rather than an analogy.

6.1. Presentations as gauges

Definition 6.1 (Presentation and load). A *presentation* of $\mathbb{Q}$ is a finite subset $P \subset \mathbb{Q}$ with $\{0, 1\} \subseteq P$. The *load* $L_P(x)$ of $x \in \mathbb{Q}$ is the least $n \geq 0$ such that x is obtained from elements of P by n applications of the binary operations $+$, $-$, $\times$, $\div$ (division by nonzero elements only). In particular, $L_P(x) = 0$ if and only if $x \in P$.

Every $x \in \mathbb{Q}$ has finite load from every presentation: integers are reached from 1 by iterated addition and one negation, and a/b by one further division. Hence all presentations generate the same field.

Definition 6.2 (Gauge moves). An *elementary gauge move* adjoins a generated element, $P \mapsto P \cup \{g\}$ with $g \in \mathbb{Q}$, or deletes a redundant primitive, $P \mapsto P \setminus \{g\}$ with $g \notin \{0, 1\}$. Two presentations are *gauge-equivalent* if connected by elementary moves; since all presentations of $\mathbb{Q}$ generate $\mathbb{Q}$, all are gauge-equivalent. A quantity is *gauge-invariant* if it depends only on the scalar x , not on the presentation used to exhibit it.

In this language, the two presentations

$$P_{\min} = \{0, 1\}, \quad P_2 = \{0, 1, 2\}$$

are two gauges, and dyadic load is a gauge-dependent quantity. The next lemma shows the dependence is total.

Lemma 6.3 (Loads are pure gauge). *For every $x \in \mathbb{Q}$ and every presentation P there is a gauge-equivalent presentation P' with $L_{P'}(x) = 0$. Consequently*

$$\inf_{P' \sim P} L_{P'}(x) = 0 \quad \text{for every } x \in \mathbb{Q},$$

and any gauge-invariant function $\ell : \mathbb{Q} \rightarrow [0, \infty)$ satisfying $\ell \leq L_{P'}$ for all gauges P' is identically zero. There is no gauge-invariant “load of x .”

Proof. Take $P' = P \cup \{x\}$; the move is elementary because x is generated by P , and $L_{P'}(x) = 0$ by definition. The remaining claims follow at once. $\square$

Lemma 6.4 (The only invariant residue of load is properness). *For every presentation P and every n , the sublevel set*

$$\{x \in \mathbb{Q} : L_P(x) \leq n\}$$

is finite. Hence for every injective sequence (x_k) in $\mathbb{Q}$ and every gauge P ,

$$L_P(x_k) \rightarrow \infty.$$

Proof. Let $E_0 = P$ and let E_{k+1} be E_k together with all outputs of one operation applied to pairs from E_k . Then

$$|E_{k+1}| \leq |E_k| + 4|E_k|^2 < \infty$$

by induction, and $\{L_P \leq n\} \subseteq E_n$. □

Lemma 6.4 is gauge-invariant but metrically empty: it declares every injective sequence heavy, so it cannot distinguish $b_n = 2^{-n}$, intuitively small in the ordinary metric, from 2^n , intuitively large. Combining the two lemmas gives the corrected reading of the dyadic example.

Corollary 6.5 (Corrected reading of the dyadic example). *Let $b_n = 2^{-n}$. Then*

$$|b_n| \rightarrow 0$$

while the dyadic load from $P_{\min}$ is n and the dyadic load from P_2 is 0. This demonstrates gauge variance of load, not a failure of the rational metric. It shows that no presentation load can serve as a notion of scalar size. If closeness is to be a property of scalars, it must be defined through the operations θ, α , invariantly.

6.2. Gauge-invariant size functions are absolute values

Loads obey the operational inequalities

$$L_P(x \circ y) \leq L_P(x) + L_P(y) + 1$$

for $\circ \in \{+, \times\}$. Exponentiating, any $s = b^{L_P}$ is submultiplicative and, after normalization, subadditive up to a constant. The gauge-invariant idealization of “size,” freed of the presentation, is therefore the following.

Definition 6.6 (Invariant size). An *invariant size* on $\mathbb{Q}$ is a function $s : \mathbb{Q} \rightarrow [0, \infty)$ such that

- (S1) $s(x) = 0$ if and only if $x = 0$ (non-degeneracy),
- (S2) $s(xy) \leq s(x)s(y)$ (compatibility with α),
- (S3) $s(x + y) \leq s(x) + s(y)$ (compatibility with θ),
- (S4) $s(x^{-1}) = s(x)^{-1}$ for $x \neq 0$ ($\alpha^\pm$ symmetry).

Axiom (S4) is the metric expression of the fact that the ladder has already adjoined multiplicative inverses: a presentation may exhibit a scalar either directly or via its inverse, and an invariant size must price the two exhibitions reciprocally.

Lemma 6.7 (Multiplicativity). *If s satisfies (S1), (S2), and (S4), then $s(1) = 1$ and*

$$s(xy) = s(x)s(y)$$

for all x, y . Hence an invariant size is exactly an absolute value on $\mathbb{Q}$.

Proof. By (S4) with $x = 1$, $s(1) = s(1)^{-1}$, so $s(1)^2 = 1$ and $s(1) = 1$. For $y \neq 0$,

$$s(x) = s((xy)y^{-1}) \leq s(xy)s(y)^{-1}$$

by (S2) and (S4). Hence $s(xy) \geq s(x)s(y)$. Together with (S2), this gives equality. The case $y = 0$ is trivial by (S1). Finally $s(-1)^2 = s(1) = 1$ gives $s(-1) = 1$, so s is an absolute value in the usual sense. $\square$

The classification of absolute values on $\mathbb{Q}$ is classical.

Theorem 6.8 (Ostrowski, 1916). *Every absolute value on $\mathbb{Q}$ is either trivial, equivalent to the Archimedean absolute value $|\cdot|_\infty$, or equivalent to a p -adic absolute value $|\cdot|_p$ for exactly one prime p . Here $s \sim s'$ means $s' = s^c$ for some $c > 0$, equivalently that s and s' induce the same topology on $\mathbb{Q}$.*

This theorem is cited, not reproved; it is the one classical input of this section. It says that the gauge-invariant sizes on $\mathbb{Q}$ form a short, exhaustive list. The elementary dyadic phenomenon sits on that list. Indeed,

$$L_{P_{\min}}^{(2)}(x) = |\nu_2(x)| = |\log_2 |x|_2|,$$

so dyadic load is the logarithmic shadow of the genuine invariant $|\cdot|_2$. The tension exhibited by $b_n = 2^{-n}$ – null for $|\cdot|_\infty$ while $|b_n|_2 = 2^n \rightarrow \infty$ – is, invariantly, the inequivalence of two absolute values. The selection problem is therefore real and must be solved by an axiom.

6.3. Generator visibility selects the Archimedean branch

The entire ladder is generated by the successor action θ from the unit 1. A size that assigns bounded values to the whole successor orbit declares the generating action metrically invisible: arbitrarily deep successor-towers would occupy a bounded region of the size scale. That is incompatible with a program in which successor-depth is the primitive notion of magnitude. We therefore impose the following axiom.

Axiom 6.9 (Generator visibility).

$$\sup_{n \in \mathbb{N}} s(\theta^n(0)) = \sup_{n \in \mathbb{N}} s(n) = \infty.$$

Lemma 6.10 (Boundedness on $\mathbb{N}$ is the ultrametric dichotomy). *For an absolute value s on $\mathbb{Q}$ the following are equivalent:*

1. $s(n) \leq 1$ for all $n \in \mathbb{N}$;
2. $s(x + y) \leq \max\{s(x), s(y)\}$ for all x, y .

Proof sketch. The implication (2) $\Rightarrow$ (1) follows by induction from $s(1) = 1$. For (1) $\Rightarrow$ (2), expand by the binomial theorem and use multiplicativity:

$$s(x + y)^n \leq \sum_{k=0}^n s \binom{n}{k} s(x)^k s(y)^{n-k} \leq (n + 1) \max\{s(x), s(y)\}^n.$$

Then take n -th roots and let $n \rightarrow \infty$. $\square$

Theorem 6.11 (Archimedean selection; uniqueness of the invariant uniformity). *Let s be an invariant size on $\mathbb{Q}$ satisfying Axiom 6.9. Then*

$$s = |\cdot|^c_\infty \quad \text{for some } c \in (0, 1].$$

All such s have the same null sequences and the same Cauchy sequences. Hence the gauge-invariant, generator-visible uniformity on $\mathbb{Q}$ exists and is unique.

Proof. By Lemma 6.7, s is an absolute value. The trivial absolute value and every $|\cdot|_p^c$ satisfy $s(n) \leq 1$ on $\mathbb{N}$; for the p -adics, $|n|_p \leq 1$ since $n \in \mathbb{Z}$. Hence they violate Axiom 6.9. Ostrowski's theorem leaves $s = |\cdot|_\infty^c$ with $c > 0$. Subadditivity (S3) at $x = y = 1$ gives $2^c = s(2) \leq 2$, so $c \leq 1$. Since $s(x_n) \rightarrow 0$ if and only if $|x_n| \rightarrow 0$, and likewise for Cauchy tails, every admissible c induces the same uniform structure. $\square$

Remark 6.12 (The dyadic shadow and the p -adic boundary). Theorem 6.11 locates the p -adic completions precisely. They are what results from keeping (S1)–(S4) and dropping generator visibility. Each $|\cdot|_p$ is a gauge-invariant size whose completion $\mathbb{Q}_p$ is a complete valued field, but one in which the successor tower is bounded. For example, the orbit $2, 4, 8, \dots$ is 2-adically null. The p -adic completions are therefore not absorbed by the Archimedean completion; they are excluded from the classical ladder by Axiom 6.9.

6.4. Incompleteness of the invariant uniformity

Proposition 6.13 ($\mathbb{Q}$ is incomplete). *There is a Cauchy sequence in $(\mathbb{Q}, |\cdot|_\infty)$ with no limit in $\mathbb{Q}$.*

Proof. Define rationals by $x_0 = 2$ and

$$x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n}, \quad e_n = x_n^2 - 2.$$

A direct computation gives

$$e_{n+1} = \left(\frac{e_n}{2x_n} \right)^2 \geq 0.$$

Moreover, $e_n > 0$ for all n : if some x_n had $x_n^2 = 2$, then $\nu_2(x_n^2) = 2\nu_2(x_n)$ would be even while $\nu_2(2) = 1$ is odd. Hence $x_n^2 > 2$, so $x_n > 1$, and

$$x_n - x_{n+1} = \frac{e_n}{2x_n} > 0.$$

Thus the sequence is strictly decreasing with $1 < x_n \leq 2$. From $x_n > 1$,

$$e_{n+1} \leq \frac{e_n^2}{4}, \quad e_1 = \frac{1}{4},$$

so $e_n \leq 4^{1-2^{n-1}}$ for $n \geq 1$, and for $m < n$,

$$0 < x_m - x_n = \sum_{k=m}^{n-1} \frac{e_k}{2x_k} \leq \frac{1}{2} \sum_{k \geq m} e_k \xrightarrow{m \rightarrow \infty} 0.$$

Thus the sequence is Cauchy in $\mathbb{Q}$. If it had a limit $q \in \mathbb{Q}$, then $q \geq 1 > 0$, and continuity of the rational operations away from 0 would give

$$q = \frac{q}{2} + \frac{1}{q},$$

so $q^2 = 2$, contradicting the valuation parity argument above. $\square$

Definition 6.14 (λ , revised). The symbol λ denotes the following obstruction and its repair: the unique gauge-invariant, generator-visible uniformity on $\mathbb{Q}$ is incomplete, and the repair is its metric completion. Presentation loads play no role in scalar identity: they are pure gauge.

7. The real numbers as the unique gauge-invariant completion

The construction of the completion is the standard Cauchy quotient. Let

$$\mathcal{C}(\mathbb{Q}) = \{(x_n) : x_n \in \mathbb{Q}, (x_n) \text{ is Cauchy}\}, \quad c_0(\mathbb{Q}) = \{(x_n) \in \mathcal{C}(\mathbb{Q}) : x_n \rightarrow 0\},$$

and define

$$\mathbb{R} = \mathcal{C}(\mathbb{Q})/c_0(\mathbb{Q}).$$

Two rational Cauchy presentations determine the same real number exactly when their difference tends to zero:

$$(x_n) \sim (y_n) \iff x_n - y_n \rightarrow 0.$$

Operations descend to the quotient:

$$[x] + [y] = [x + y], \quad [x][y] = [xy], \quad -[x] = [-x].$$

The embedding of rationals is

$$\iota : \mathbb{Q} \hookrightarrow \mathbb{R}, \quad q \mapsto [(q, q, q, \dots)].$$

The quotient is a field. If $[x] \neq 0$, then x is not a null sequence. Since x is Cauchy, it is eventually bounded away from zero: there are $\varepsilon > 0$ and N such that

$$n > N \implies |x_n| > \varepsilon.$$

Define a rational Cauchy sequence y by

$$y_n = \begin{cases} 1, & n \leq N, \\ 1/x_n, & n > N. \end{cases}$$

Then $xy - 1 \in c_0(\mathbb{Q})$, so $[x][y] = [1]$. Hence every nonzero real class has an inverse.

The ordered-field structure is obtained by declaring $[x] > 0$ when some representative is eventually bounded below by a positive rational:

$$\exists \varepsilon \in \mathbb{Q}_{>0} \exists N \forall n > N, \quad x_n > \varepsilon.$$

The construction yields the complete Archimedean ordered field.

What is new in the derivation above is that the completion is not merely chosen. It is forced, and forced uniquely.

Theorem 7.1 (Necessity and uniqueness of $\mathbb{R}$). *Let $K \supseteq \mathbb{Q}$ be any scalar system such that*

- (i) *the gauge-invariant, generator-visible size of Theorem 6.11 extends to an absolute value on K ;*
- (ii) *every Cauchy sequence of K converges in K ;*
- (iii) *K is minimal with (i) and (ii), i.e. no proper subfield containing $\mathbb{Q}$ satisfies both.*

Then K is isomorphic, by a unique isomorphism fixing $\mathbb{Q}$ and preserving the size, to

$$\mathbb{R} = \mathcal{C}(\mathbb{Q})/c_0(\mathbb{Q}).$$

Moreover, $\mathbb{R}$ is, up to unique isomorphism, the only complete Archimedean ordered field, and it satisfies the universal property: every uniformly continuous map from $\mathbb{Q}$ into a complete metric space extends uniquely to $\mathbb{R}$.

Proof sketch. The closure of $\mathbb{Q}$ in K is a completion of $(\mathbb{Q}, |\cdot|)$; by minimality it is all of K . Any two completions of a metric space are uniquely isometric over the space, and here the isometry is a field isomorphism because the operations are uniformly continuous on bounded sets and determined by their restrictions to the dense subfield $\mathbb{Q}$. Completeness plus the Archimedean property of the extended absolute value yields the complete Archimedean ordered field, whose uniqueness up to unique isomorphism is classical. The extension property is the universal property of metric completion. $\square$

Remark 7.2 (The λ chain, displayed). The transition $\mathbb{Q} \rightarrow \mathbb{R}$ is the composite of forced steps:

$$\underbrace{\text{loads are pure gauge}}_{\text{Lemma 6.3}} \implies \underbrace{\text{size must be an invariant absolute value}}_{\text{Lemma 6.7}} \implies \underbrace{\{|\cdot|_\infty\} \cup \{|\cdot|_p\}}_{\text{Ostrowski}} \implies \underbrace{|\cdot|_\infty \text{ only}}_{\text{Axiom 6.9}} \implies \underbrace{\mathbb{R}}_{\text{Proposition 6.10}}$$

In compact notation, $\mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda)$ as before, but λ is no longer an interpretive label. It is the statement that gauge variance of presentation load, once reduced to its invariant content, admits exactly one generator-visible uniformity, and that uniformity is incomplete over $\mathbb{Q}$.

8. Exponentiation as a coupled additive-multiplicative action

After $\mathbb{R}$ is obtained, both addition and multiplication are reversible in the usual scalar sense, and metric completion is available. Yet a remaining obstruction appears when additive parameters are coupled to multiplicative action by exponentiation.

Real exponentiation gives

$$\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}, \quad \exp(t) = e^t.$$

This is a one-parameter multiplicative subgroup of positive real scalars:

$$e^{s+t} = e^s e^t.$$

However, it is line-bound:

$$e^t > 0 \quad \text{for all } t \in \mathbb{R}.$$

It scales. It does not rotate.

If the scalar system is required to support a magnitude-preserving polar one-parameter action, the real line is insufficient. One needs an element whose exponential generates rotation rather than positive scaling.

This is the obstruction labelled ρ in this note.

9. The complex numbers: polar complexification

Definition 9.1 (Complexification of $\mathbb{R}$). Adjoin an element i satisfying

$$i^2 = -1.$$

Equivalently,

$$\mathbb{C} = \mathbb{R}[x]/(x^2 + 1).$$

Every element has the form

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

As a real vector space,

$$\mathbb{C} \cong \mathbb{R}^2.$$

Multiplication is

$$(a, b)(c, d) = (ac - bd, ad + bc).$$

The polar unit repairs the exponentiation obstruction:

$$e^{it} = \cos t + i \sin t.$$

This satisfies

$$|e^{it}| = 1,$$

and therefore gives a magnitude-preserving rotation action:

$$z \mapsto e^{it} z.$$

Thus the transition

$$\mathbb{R} \longrightarrow \mathbb{C}$$

is not only the algebraic solution of

$$x^2 + 1 = 0.$$

That equation is the minimal polynomial symptom. The structural repair is polar complexification: real scaling is extended to include unit-magnitude rotation.

In compact notation,

$$\boxed{\mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).}$$

The standard algebraic fact behind the terminal role of $\mathbb{C}$ is that $\mathbb{C}$ is algebraically closed: every nonconstant polynomial with complex coefficients has a complex root. This fact is used here only inside the claim boundary stated below.

10. The full scalar ladder

The complete construction is

$$\boxed{0 \longrightarrow S_{\min} \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}.}$$

Equivalently,

$$\boxed{0 \longrightarrow \{0, 1\} \longrightarrow \theta^+ \longrightarrow \theta^\pm \longrightarrow \alpha^\pm \longrightarrow \lambda \longrightarrow \rho.}$$

System	Available structure	Construction added	Public reading
0	no scalar distinction	none	null starting stage
$S_{\min}$	0, 1	minimal generating set	identity and unit generator
$\mathbb{N}$	θ^+, α^+	successor closure	forward additive generation; semiring
$\mathbb{Z}$	$\theta^\pm, \alpha^+$	group completion	additive inverses; ring
$\mathbb{Q}$	$\theta^\pm, \alpha^\pm$	localization	multiplicative inverses; field
$\mathbb{R}$	$\theta^\pm, \alpha^\pm, \lambda$	selected metric completion	gauge-invariant, generator-visible completion; complete ordered field
$\mathbb{C}$	$\theta^\pm, \alpha^\pm, \lambda, \rho$	complexification	polar unit; algebraically closed field

11. Universal properties

Each transition can be stated by a universal property.

$\mathbb{N}$

The additive monoid $(\mathbb{N}, +, 0)$ is the free commutative monoid on one generator. For every commutative monoid M and every element $m \in M$, there is a unique monoid homomorphism

$$f : \mathbb{N} \rightarrow M$$

with $f(1) = m$.

$\mathbb{Z}$

The integers are the group completion of the additive monoid $\mathbb{N}$. For every abelian group G and monoid homomorphism

$$f : \mathbb{N} \rightarrow G,$$

there is a unique group homomorphism

$$\bar{f} : \mathbb{Z} \rightarrow G$$

such that f factors through $\mathbb{Z}$.

$\mathbb{Q}$

The rationals are the localization of $\mathbb{Z}$ at the nonzero integers. For every ring R and ring homomorphism

$$f : \mathbb{Z} \rightarrow R$$

that sends every nonzero integer to a unit of R , there is a unique ring homomorphism

$$\bar{f} : \mathbb{Q} \rightarrow R$$

through which f factors.

$\mathbb{R}$

The reals are the metric completion of $\mathbb{Q}$ under the unique gauge-invariant, generator-visible uniformity selected in Theorem 6.11. Every uniformly continuous map from $\mathbb{Q}$ into a complete metric space extends uniquely to $\mathbb{R}$ when the required hypotheses hold. The completion is derived from gauge invariance, generator visibility, and Ostrowski's classification, not merely motivated by a dyadic example.

$\mathbb{C}$

The complex numbers are the quadratic extension

$$\mathbb{R}[x]/(x^2 + 1)$$

and the algebraic closure of $\mathbb{R}$. They also supply the one-dimensional polar unitary group

$$U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{it} : t \in \mathbb{R}\}.$$

12. Terminal scalar postulate

Postulate 12.1 (Terminal classical scalar postulate). *Subject to the following hypotheses:*

1. the system is generated from $S_{\min} = \{0, 1\}$;

2. *addition and multiplication are commutative scalar operations with identities;*
 3. *additive inverses are adjoined by group completion;*
 4. *multiplicative inverses for nonzero scalars are adjoined by localization;*
 5. *scalar size is gauge-invariant in the sense of (S1)–(S4), generator-visible in the sense of Axiom 6.9, and Cauchy convergence has endpoints;*
 6. *exponentiation is extended to include magnitude-preserving polar rotation;*
 7. *equivalence is taken up to isomorphism or harmless change of presentation;*
- then $\mathbb{C}$ is the terminal classical scalar layer forced by these requirements.

This postulate does not claim that no other algebraic systems exist. It does not exclude finite fields, p -adic fields, nonstandard real fields, surreal numbers, quaternions, Clifford algebras, matrix algebras, operator algebras, schemes, sheaves, or higher categories. Those systems either alter the hypotheses, change the notion of scalar, abandon commutativity, use a different absolute value, add dimension rather than scalar closure, or move into a different category.

The postulate claims only that the ladder

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$$

is terminal for the stated classical scalar program.

13. Open questions and proof obligations

The terminal postulate is intentionally strong. A complete proof requires at least the following work.

1. **Category of presentations.** Define the category whose objects are scalar presentations and whose morphisms preserve 0, 1, addition, multiplication, completion data, and polar structure.
2. **Load invariance beyond $\mathbb{Q}$.** For $\mathbb{Q}$, pointwise presentation loads are pure gauge and the invariant size functions are absolute values. The general problem is to characterize the analogous invariant load-like functions for broader scalar presentations.
3. **p -adic branches.** The p -adic completions are not absorbed by the Archimedean completion. They are excluded here by generator visibility. A broader theory should classify the branch obtained when generator visibility is weakened or replaced.
4. **Equivalence of real completions.** Relate the Cauchy construction, Dedekind cuts, nested intervals, and other constructions of $\mathbb{R}$ inside the presentation-size argument.
5. **Polarization criterion.** Specify the exact universal property that forces $\mathbb{C}$ from $\mathbb{R}$: algebraic closure, polar exponentiation, two-dimensional real rotation, or a combined condition.
6. **Boundary systems.** Classify precisely why p -adics, quaternions, nonstandard reals, surreal numbers, and operator algebras fall outside the terminal scalar postulate or require a modified postulate.
7. **Terminality theorem.** Convert the postulate into a theorem by proving that any scalar system satisfying the hypotheses factors through $\mathbb{C}$ uniquely up to the chosen equivalence.

14. Final compression

The derivation can be compressed to one line:

$$\boxed{\mathbb{C} = S_{\min} + \theta^{\pm} + \alpha^{\pm} + \lambda + \rho.}$$

Expanded in public language:

$\mathbb{C}$ = minimal generating set
 + additive inverses
 + multiplicative inverses
 + gauge-invariant, generator-visible completion
 + polar complexification.

Thus the number systems are not presented as a list of unrelated enlargements. They are a sequence of forced repairs to partial scalar structure:

$\mathbb{N}$ generates; $\mathbb{Z}$ reverses addition; $\mathbb{Q}$ reverses multiplication;
 $\mathbb{R}$ completes the selected invariant uniformity; $\mathbb{C}$ polarizes exponentiation.

References

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